



Mark Scheme (Results)

October 2025

Pearson Edexcel International Advanced
Level in Chemistry
WCH14/01A

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General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.

Using the Mark Scheme

Examiners should look for qualities to reward rather than faults to penalise. This does NOT mean giving credit for incorrect or inadequate answers, but it does mean allowing candidates to be rewarded for answers showing correct application of principles and knowledge. Examiners should therefore read carefully and consider every response: even if it is not what is expected it may be worthy of credit.

The mark scheme gives examiners:

- an idea of the types of response expected
- how individual marks are to be awarded
- the total mark for each question
- examples of responses that should NOT receive credit.

/ means that the responses are alternatives and either answer should receive full credit.

() means that a phrase/word is not essential for the award of the mark, but helps the examiner to get the sense of the expected answer.

Phrases/words in **bold** indicate that the meaning of the phrase or the actual word is **essential** to the answer.

ecf/TE/cq (error carried forward) means that a wrong answer given in an earlier part of a question is used correctly in answer to a later part of the same question.

Candidates must make their meaning clear to the examiner to gain the mark. Make sure that the answer makes sense. Do not give credit for correct words/phrases which are put together in a meaningless manner. Answers must be in the correct context.

Quality of Written Communication

Questions which involve the writing of continuous prose will expect candidates to:

- write legibly, with accurate use of spelling, grammar and punctuation in order to make the meaning clear
- select and use a form and style of writing appropriate to purpose and to complex subject matter
- organise information clearly and coherently, using specialist vocabulary when appropriate.

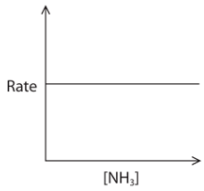
Full marks will be awarded if the candidate has demonstrated the above abilities. Questions where QWC is likely to be particularly important are indicated (QWC) in the mark scheme, but this does not preclude others.

Section A

Question Number	Answer	Mark
1	The only correct answer is B (electrical conductivity) <i>A is incorrect because none of the reactants or products are coloured</i> <i>C is incorrect because there is no change in mass</i> <i>D is incorrect because titration is not a continuous monitoring method</i>	(1)

Question Number	Answer	Mark
2(a)	The only correct answer is A (rate = k) <i>B is incorrect because this is a first order rate equation</i> <i>C is incorrect because this is a second order rate equation</i> <i>D is incorrect because this shows overall order four and refers to the reverse reaction</i>	(1)

Question Number	Answer	Mark
2(b)	The only correct answer is C ($\text{mol dm}^{-3} \text{ s}^{-1}$) <i>A is incorrect because rate constants always have units</i> <i>B is incorrect because these are the units of a first order rate constant</i> <i>D is incorrect because these are the units of a second order rate constant</i>	(1)

Question Number	Answer	Mark
2(c)	 The only correct answer is A () <i>B is incorrect because this graph shows a first order reaction</i> <i>C is incorrect because this graph shows a first order reaction</i> <i>D is incorrect because this graph shows no reaction occurring</i>	(1)

Question Number	Answer	Mark
3	The only correct answer is D (S_N1 ; two) <i>A is incorrect because the halogenoalkane is tertiary so the mechanism would be S_N1 which has two steps</i> <i>B is incorrect because the halogenoalkane is tertiary and so the mechanism would be S_N1</i> <i>C is incorrect because the S_N1 mechanism has two steps</i>	(1)

Question Number	Answer	Mark
4	$K_c = \frac{[\text{CO}][\text{H}_2]}{[\text{H}_2\text{O}]}$ The only correct answer is C () <i>A is incorrect because the concentration of steam has been omitted</i> <i>B is incorrect because the concentration of steam has been omitted and carbon is not in the gas phase</i> <i>D is incorrect because carbon is not in the gas phase</i>	(1)

Question Number	Answer	Mark
5	The only correct answer is C (CH_2ClCOOH and $\text{CH}_3\text{COOH}_2^+$) <i>A is incorrect because ethanoic acid accepts a proton in this system so is a base</i> <i>B is incorrect because ethanoic acid accepts a proton in this system so is a base</i> <i>D is incorrect because $\text{CH}_2\text{ClCOO}^-$ is a base</i>	(1)

Question Number	Answer	Mark
6	The only correct answer is B (13.0) <i>A is incorrect because the concentration of hydroxide ions is not 0.20 mol dm^{-3}</i> <i>C is incorrect because the concentration of hydroxide ions is not $0.050 \text{ mol dm}^{-3}$</i> <i>D is incorrect because the concentration of hydroxide ions is not $0.025 \text{ mol dm}^{-3}$</i>	(1)

Question Number	Answer	Mark
7	The only correct answer is A (5.06) <i>B is incorrect because this is the pH of an equimolar solution of CH_3COOH and CH_3COONa</i> <i>C is incorrect because this is the pH when the concentrations are reversed</i> <i>D is incorrect because this is the pH of a $0.100 \text{ mol dm}^{-3}$ solution of CH_3COOH</i>	(1)

Question Number	Answer	Mark
8	The only correct answer is B (bromocresol green ($\text{pK}_{\text{in}} = 4.7$)) <i>A is incorrect because the pH range would not lie within the vertical section of a strong acid and weak base titration</i> <i>C is incorrect because the pH range would not lie within the vertical section of a strong acid and weak base titration</i> <i>D is incorrect because the pH range would not lie within the vertical section of a strong acid and weak base titration</i>	(1)

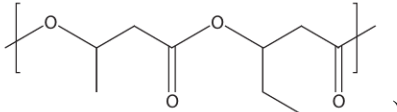
Question Number	Answer	Mark
9(a)	The only correct answer is C (the second ionisation energy of strontium is $+1614 \text{ kJ mol}^{-1}$) <i>A is incorrect because lattice energies are always exothermic</i> <i>B is incorrect because -590 is twice the first electron affinity of iodine</i> <i>C is incorrect because the standard enthalpy change of atomisation of strontium is $+164 \text{ kJ mol}^{-1}$</i>	(1)

Question Number	Answer	Mark
9(b)	The only correct answer is D (+107) <i>A is incorrect because the sign of the enthalpy change of atomisation of strontium has not been reversed and the answer has not been divided by two</i> <i>B is incorrect because the sign of the enthalpy change of atomisation of strontium has not been reversed</i> <i>C is incorrect because +214 is the enthalpy change in producing 2 mol of I(g)</i>	(1)

Question Number	Answer	Mark
10(a)	The only correct answer is B (retention time) <i>A is incorrect because this term does not relate to gas chromatography</i> <i>C is incorrect because this term is not used</i> <i>D is incorrect because this term is not used</i>	(1)

Question Number	Answer	Mark
10(b)	The only correct answer is A (argon) <i>B is incorrect because the carrier gas must be inert</i> <i>C is incorrect because the carrier gas must be inert</i> <i>D is incorrect because the carrier gas must be inert</i>	(1)

Question Number	Answer	Mark
10(c)	The only correct answer is B (84 mm) <i>A is incorrect because 840 mm is a factor of 10 too large</i> <i>C is incorrect because this is the distance moved by the amino acid</i> <i>D is incorrect because this is the distance moved multiplied by the distance travelled by the amino acid</i>	(1)

Question Number	Answer	Mark
11	 The only correct answer is A (<i>B is incorrect because there is an extra carbon atom at the left hand end of the repeat unit</i> <i>C is incorrect because this repeat unit has an extra oxygen and the structure is incorrect</i> <i>D is incorrect because this repeat unit has an extra oxygen</i>	(1)

Question Number	Answer	Mark
12	The only correct answer is D (CH₃COCl) <i>A is incorrect because the reaction with a ketone would not produce an N-substituted amide</i> <i>B is incorrect because any reaction with the carboxylic acid would be too slow at room temperature</i> <i>C is incorrect because any reaction with the ester would be too slow at room temperature</i>	(1)

Question Number	Answer	Mark
13(a)	The only correct answer is C (Y and Z only) <i>A is incorrect because neither W nor X are methyl ketones</i> <i>B is incorrect because Y is not the only molecule which is a methyl ketone</i> <i>D is incorrect because Z is not the only molecule which is a methyl ketone</i>	(1)

Question Number	Answer	Mark
13(b)	The only correct answer is C (W and X only) <i>A is incorrect because X also gives a positive result</i> <i>B is incorrect because W also gives a positive result</i> <i>D is incorrect because neither Y nor Z gives a positive result</i>	(1)

Question Number	Answer	Mark
13(c)	The only correct answer is D (W, X and Y only) <i>A is incorrect because Y has a CH₃CH₂ group likely to produce a mass spectrum fragment with a $m/e = 29$</i> <i>B is incorrect because W has a HCO group likely to produce a mass spectrum fragment with a $m/e = 29$</i> <i>C is incorrect because Z would not be expected to produce a mass spectrum fragment with a $m/e = 29$</i>	(1)

TOTAL FOR SECTION A = 20 MARKS

Section B

Question Number	Answer	Additional Guidance	Mark
14(a)(i)	$K_p = \frac{p(\text{C}_2\text{H}_5\text{OH})}{p(\text{C}_2\text{H}_4) \times p(\text{H}_2\text{O})}$	Allow without brackets Allow pp for p Allow p inside brackets Do not award square brackets Do not award omission of 'p' Penalise omission of K_p	(1)

Question Number	Answer	Additional Guidance	Mark																												
14(a)(ii)	An answer that makes reference to the following points: - calculation of moles at equilibrium (1) - calculation of mole fractions at equilibrium (1) - calculation of partial pressures at equilibrium (1) - substitution of values into equation and calculation of K_p (1) - units (1)	<u>Example of calculation</u> <table><tr><td></td><td>Ethene</td><td>Steam</td><td>Ethanol</td></tr><tr><td>Initial moles</td><td>1.00</td><td>1.00</td><td>0.00</td></tr><tr><td>Equilibrium moles</td><td>0.570</td><td>0.570</td><td>0.430</td></tr><tr><td>Mole fraction (MF)</td><td>0.363</td><td>0.363</td><td>0.274</td></tr><tr><td>Partial pressure (MF x 50.0)</td><td>18.15(3)</td><td>18.15(3)</td><td>13.69(4)</td></tr><tr><td>$K_p = (13.694 \div 18.153^2) =$</td><td colspan="3">0.0416 / 0.041557</td></tr><tr><td colspan="4">atm⁻¹</td></tr></table> TE on expression in (a)(i) and at each stage Ignore SF except 1SF Correct answer with some working scores (5)		Ethene	Steam	Ethanol	Initial moles	1.00	1.00	0.00	Equilibrium moles	0.570	0.570	0.430	Mole fraction (MF)	0.363	0.363	0.274	Partial pressure (MF x 50.0)	18.15(3)	18.15(3)	13.69(4)	$K_p = (13.694 \div 18.153^2) =$	0.0416 / 0.041557			atm ⁻¹				(5)
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Question Number	Answer	Additional Guidance	Mark
14(b)	An answer that makes reference to the following points: - high pressure favours the formation of ethanol and because 2 mol (of gas from 1 mol) - high temperature does not favour the formation of ethanol and because the (forward) reaction is exothermic - high temperature is needed because otherwise the rate of reaction is too slow	Allow reverse arguments Ignore comments about K_c / K_p Allow high pressure increases yield (of ethanol) Allow reactions shifts to the right at high temperature (1) Allow fewer moles on the right hand side Allow high temperature decreases yield (of ethanol) Allow reaction shifts to the left at high temperature (1) Accept because the reverse reaction is endothermic (1) Allow unconverted reactants can be recycled to increase the overall yield Ignore higher temperature increases rate If no other mark is scored, Allow (1) for: High pressure increases yield / causes equilibrium to shift to right and High temperature reduces yield / causes equilibrium to shift to left	(3)

(Total for Question 14 = 9 marks)

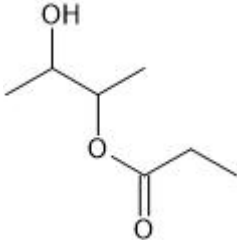
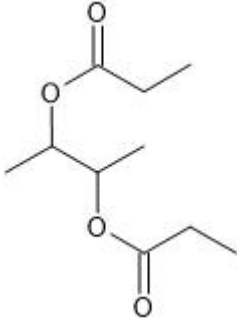
Question Number	Answer	Additional Guidance	Mark
15(a)	An answer that makes reference to the following points: - balancing of reducing agent (1) - butane-2,3-diol structure (1)	Example of equation $\text{CH}_3\text{COCOCH}_3 + 4[\text{H}] \rightarrow \text{CH}_3\text{CHOHCHOHCH}_3$ Do not award if any other additional product given Allow any unambiguous structure- skeletal/displayed/hybrid Do not award $\text{CH}_3\text{COH}_2\text{COH}_2\text{CH}_3$	(2)

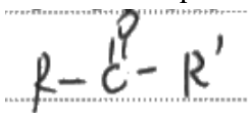
Question Number	Answer	Additional Guidance	Mark
15(b)	An answer that makes reference to the following point: go colourless / yellow-green colour disappears		(1)

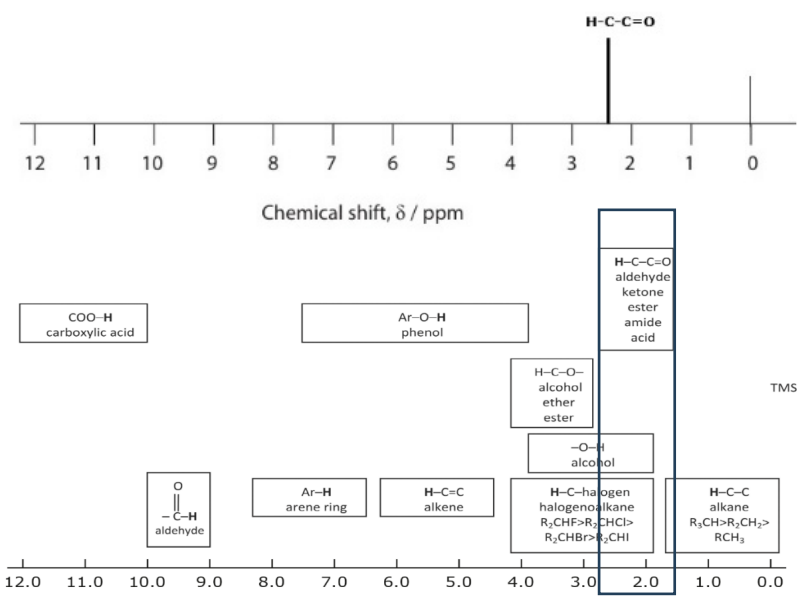
Question Number	Answer	Additional Guidance	Mark
15(c)	An answer that makes reference to the following points: - butane-2,3-diol (has the highest boiling temperature) and because (only) it has hydrogen bonding (1) - which requires more energy to break (1)	Reference to butane-2,3-dione having the higher boiling temperature scores (0) Ignore hydrogen bonding is the strongest intermolecular force Ignore hydrogen bonds are stronger than other intermolecular forces Do not award any implication that covalent bonds within butane-2,3-diol are broken	(2)

Question Number	Answer	Additional Guidance	Mark
15(d)	An answer that makes reference to the following points: - optical isomerism (1) - because of the presence of (two) chiral carbon(s) or because it forms non-superimposable mirror images (1)	Allow optical isomer Allow it is optically active Ignore the number of isomers if given Allow butane-2,3-diol is (a) chiral (molecule) Allow because of the presence of a carbon with four different groups/atoms attached / asymmetric carbon atom(s) Ignore just reference to enantiomers Ignore references to rotation of (plane) polarised light	(2) \

Question Number	Answer	Additional Guidance	Mark
15(e)(i)	An answer that makes reference to the following points: - the reaction is irreversible (not reversible) (1) - by-product is hydrogen chloride not water (1) - the reaction is fast / occurs at room temperature so no acid catalyst needed (1)	Accept reverse arguments Allow reaction (with propanoyl chloride) goes to completion Allow (propanoyl chloride) gives a higher yield Allow misty / steamy fumes for HCl Allow vigorous for fast Ignore comments about differences in reaction mechanisms/atom economy	(3)

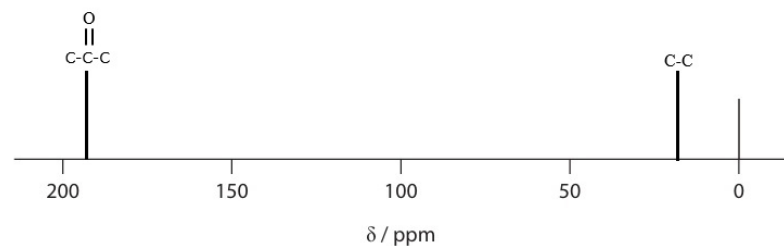
Question Number	Answer	Additional Guidance	Mark
15(e)(ii)	- skeletal formula with one ester group - skeletal formula with two ester groups	(1)  (1)  Diester skeletal formula scores (2) Displayed /structural/hybrid formulae for the diester scores (1) Penalise extra -CH₂- groups once only Penalise polymer brackets once only	(2)

Question Number	Answer	Additional Guidance	Mark
15(f)	An answer that makes reference to the following points: - butanedione has a C=O absorption at 1720–1700 (cm⁻¹ but butane-2,3-diol does not) (1) - butane-2,3-diol has a O–H absorption at 3750–3200 (cm⁻¹ but butane-2,3-diol does not) (1)	For both, allow single numbers within the quoted ranges Do not award –C=O or  Allow OH bond but not just OH Do not award -O-H Two correct wavenumber ranges without bonds scores (1) Ignore C-H absorptions Ignore C-O in the diol but NOT in the dione Penalise additional incorrect absorptions once only.	(2)

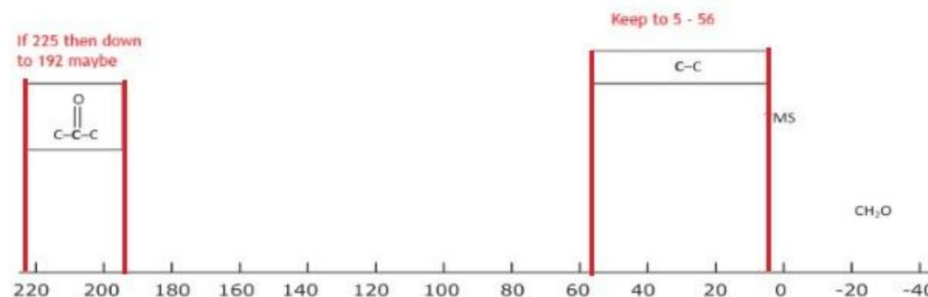
Question Number	Answer	Additional Guidance	Mark
15(g)	^1H spectrum with one singlet peak or line (1)	Accept labels for peaks drawn on butane-2,3-dione Penalise missing labels once only Ignore shape of peak / line Ignore height of lines Ignore peak area, even if incorrect <u>Example of H-1 spectra</u>  Accept a single peak / line in the range 1.6 – 2.8 Do not award any splitting of line	(3)

- ^{13}C spectrum peak / line for carbonyl carbon (1)
- ^{13}C spectrum peak / line for alkyl carbon (1)

Example of C-13 spectra



^{13}C nuclear magnetic resonance chemical shifts relative to tetramethylsilane (TMS)



Accept a peak / line in the range of 195 - 225 for carbonyl carbon

Accept a peak / line in the range of 5 – 56 for alkyl carbon

Penalise splitting of C-13 peaks once only if not penalised in H-1 spectrum

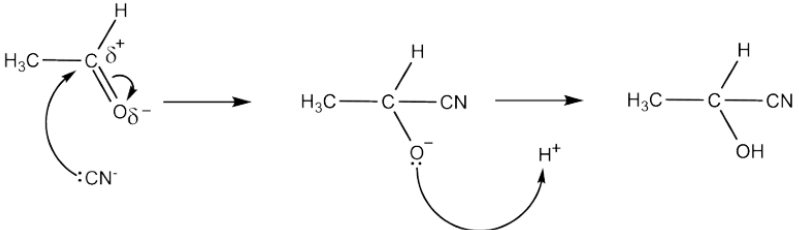
(Total for Question 15 = 17 marks)

Question Number	Answer	Additional Guidance	Mark																				
*16	This question assesses the student’s ability to show a coherent and logically structured answer with linkages and fully sustained reasoning. Marks are awarded for indicative content and for how the answer is structured and shows lines of reasoning. The following table shows how the marks should be awarded for indicative content. <table><tr><th>Number of indicative marking points seen in answer</th><th>Number of marks awarded for indicative marking points</th></tr><tr><td>6</td><td>4</td></tr><tr><td>5-4</td><td>3</td></tr><tr><td>3-2</td><td>2</td></tr><tr><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td></tr></table> The following table shows how the marks should be awarded for structure and lines of reasoning <table><tr><th></th><th>Number of marks awarded for structure of answer and sustained lines of reasoning</th></tr><tr><td>Answer shows a coherent logical structure with linkages and fully sustained lines of reasoning demonstrated throughout</td><td>2</td></tr><tr><td>Answer is partially structured with some linkages and lines of reasoning</td><td>1</td></tr><tr><td>Answer has no linkages between points and is unstructured</td><td>0</td></tr></table>	Number of indicative marking points seen in answer	Number of marks awarded for indicative marking points	6	4	5-4	3	3-2	2	1	1	0	0		Number of marks awarded for structure of answer and sustained lines of reasoning	Answer shows a coherent logical structure with linkages and fully sustained lines of reasoning demonstrated throughout	2	Answer is partially structured with some linkages and lines of reasoning	1	Answer has no linkages between points and is unstructured	0	Guidance on how the mark scheme should be applied. The mark for indicative content should be added to the mark for lines of reasoning. For example, a response with five indicative marking points that is partially structured with some linkages and lines of reasoning scores 4 marks (3 marks for indicative content and 1 mark for partial structure and some linkages and lines of reasoning). If there were no linkages between the points, then the same indicative marking points would yield an overall score of 3 marks (3 marks for indicative content and no marks for linkages). In general it would be expected that 5 or 6 indicative points would get 2 reasoning marks 3 or 4 indicative points would get 1 reasoning mark 0, 1 or 2 indicative points would get zero reasoning marks If there is any incorrect chemistry, deduct mark(s) from the reasoning. If no reasoning mark(s) awarded do not deduct mark(s). Comment: Look for the indicative marking points first, then consider the mark for the structure of the answer and sustained line of reasoning Allow reverse arguments throughout	(6)
Number of indicative marking points seen in answer	Number of marks awarded for indicative marking points																						
6	4																						
5-4	3																						
3-2	2																						
1	1																						
0	0																						
	Number of marks awarded for structure of answer and sustained lines of reasoning																						
Answer shows a coherent logical structure with linkages and fully sustained lines of reasoning demonstrated throughout	2																						
Answer is partially structured with some linkages and lines of reasoning	1																						
Answer has no linkages between points and is unstructured	0																						

	Indicative content IP1 the bonding in potassium chloride is (almost) 100% ionic and as the theoretical and Born-Haber values are (very) similar IP2 the bonding in calcium iodide has some covalent character and as the theoretical and Born-Haber values are different IP3 the anion/iodide ion in calcium iodide is more polarised (than in potassium chloride) IP4 the calcium ion has greater charge density so greater polarising power IP5 the iodide ion is larger (than the chloride ion and so is more easily polarised) IP6 the bonding in calcium iodide is stronger than the bonding in potassium chloride because the charge on the calcium ion is twice as large (as the charge on the potassium ion)	Penalise molecules once in whole answer. Allow calculated difference in values for IP1 & IP2 Allow IP1 and IP2 if bonding in KCl is 100% ionic and CaI₂ has covalent character, and difference is larger in CaI₂ (than in KCl) If IP1 and IP2 not scored then allow 1IP for bonding in KCl is 100% ionic and CaI₂ has covalent character Allow even if comparing CaCl₂ with KCl providing it is clear that it is the anion being polarised. Can be inferred from an IP5 statement so long as linked to larger size of Ca²⁺ Ignore CaI₂ is more polarised (than KCl) Allow the calcium ion has greater charge / is smaller than a potassium ion so has greater polarising power Ignore references to size of an anion Polarisation must be mentioned at least once in IP3 or IP4. Penalise lack of comparative language one only in IP4, IP5 & IP6 Allow if Ca²⁺ and K⁺ are seen anywhere in answer Allow calcium iodide has stronger bonding (than expected) due to having covalent character. Penalise reference to atoms instead of ions once only in IP3 to IP5.	
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(Total for Question 6 = 6 marks)

Question Number	Answer	Additional Guidance	Mark
17(a)	An answer that makes reference to the following point 2-hydroxypropanoic acid	Ignore punctuation Allow 2-hydroxylpropanoic acid	(1)

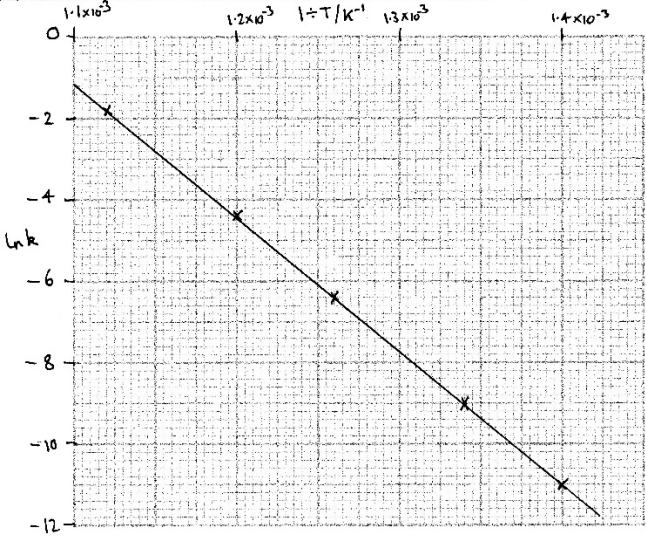
Question Number	Answer	Additional Guidance	Mark
17(b)(i)	- dipole on C=O bond - curly arrow from C=O bond to O or just beyond - lone pair on the C of cyanide ion - curly arrow from C of cyanide ion to C of C=O - structure of intermediate, including charge - lone pair on O of intermediate - H⁺ or H^{δ+} of HCN/H₂O - curly arrow from O to H⁺/HCN/H₂O 8 points scores (4) 6 – 7 scores (3) 4 – 5 scores (2) 2 – 3 scores (1)	<u>Example of mechanism</u>  Lone pair must be wholly placed on the C If lone pair placed on N of CN⁻, allow curly arrow from this lone pair If curly arrow goes to H of HCN / H₂O there must be a 2nd arrow from H-C bond to C / H-O bond to O If lone pair present for BP 3 and/or BP 6, curly arrow must come from lone pair. If lone pair not present for BP 3 and/or BP6, curly arrow may come from anywhere on C or O.	(4)

Question Number	Answer	Additional Guidance	Mark
17(b)(ii)	An answer that makes reference to the following point: hydrolysis	Allow acid / acidic hydrolysis Allow base / basic hydrolysis Allow alkali / alkaline hydrolysis Do not award hydration / dehydration / substitution	(1)

Question Number	Answer	Additional Guidance	Mark
17(c)	correct equation	Accept displayed/structural/skeletal formulae or combination thereof <u>Example of equation</u> $ \begin{array}{c} \text{OH} \\ \\ \text{H}_3\text{C}-\text{C}-\text{C} \\ \quad \diagup \\ \text{H} \quad \text{O} \\ \quad \quad \quad \diagdown \\ \quad \quad \quad \text{OH} \end{array} + \text{NaHCO}_3 \longrightarrow \begin{array}{c} \text{OH} \\ \\ \text{H}_3\text{C}-\text{C}-\text{C} \\ \quad \diagup \\ \text{H} \quad \text{O} \\ \quad \quad \quad \diagdown \\ \quad \quad \quad \text{O}^- \text{Na}^+ \end{array} + \text{CO}_2 + \text{H}_2\text{O} $ Ignore state symbols even if incorrect Allow H₂CO₃ Allow without charges on O⁻Na⁺ Do not award covalent bond drawn between O and Na	(1)

(Total for Question 17 = 7 marks)

Question Number	Answer	Additional Guidance	Mark
18(a)(i)	$(1 \div T =) 1.26 \times 10^{-3}$ - and $(\ln k =) -4.40$	1) Both values must be to 2DP 1)	(1)

Question Number	Answer	Additional Guidance	Mark
18(a)(ii)	- axes correct way round and labelled - suitable scale with points covering at least half the axes in both directions - all points plotted ± 1 small square and straight line	(1) Do not award t for T Do not award $\ln K$ scale inverted (ie 0 at bottom, -12 at top) (1) Allow $\times 10^{-3}$ to be to be part of the units on the x-axis (1) <u>Example of graph</u> 	(3)

Question Number	Answer	Additional Guidance	Mark
18(a)(iii)	• calculation of gradient of line with sign • K unit for gradient • calculation of activation energy	(1) Gradient range -32000 to -33700 Can be calculated using the points in the table (1) No TE for incorrect units on graph (1) <u>Example of calculation</u> $E_a = (-32857 \times -8.31 \div 1000) = 273 \text{ (kJ mol}^{-1}\text{)}$ E_a range 265.9 to 280.0 Allow final answer in J mol^{-1} but units must be stated TE from incorrect gradient but E_a must be positive Ignore SF except for 1SF Do not award negative E_a	(3)

Question Number	Answer	Additional Guidance	Mark
18(b)	An answer that makes reference to the following points: - rate constant k increases with increasing temperature (1) - so a greater proportion of collisions / molecules / particles have energy greater or equal to the activation energy (1) - so the frequency of successful collisions increases or more successful collisions per unit time (1)	Allow $\ln k$ increases Ignore reference to $1/T$ Ignore any reference to equilibrium constant Allow more particles have enough energy to react Allow more for greater proportion of Allow effective for successful	(3)

(Total for Question 18 = 10 marks)
TOTAL FOR SECTION B = 49 MARKS

Section C

Question Number	Answer	Additional Guidance	Mark
19(a)(i)	An answer that makes reference to two of the following points: • bubbles / fizzing / effervescence • liquid / paste produced • temperature decreases • pungent smell (of ammonia)	Apply list principle (1) Ignore gas formed (1) Allow solid(s) turn to liquid(s) Ignore solid disappears / decreases Do not award (white) precipitate (1) Do not award smell of rotten eggs Ignore chemical tests for water and ammonia	(2)

Question Number	Answer	Additional Guidance	Mark
19(a)(ii)	An answer that makes reference to the following points: • (high) positive value expected and because a gas is formed (from (two) solid(s)) • because increase in moles / 3 mol to 13 mol	Allow $\Delta S_{\text{sys}} > 0$ Ignore ΔS_{sys} increases (1) Allow because liquid formed from (two) solid(s) (1) If number of moles quoted they must be correct Allow more moles of gas on RHS Allow molecules / particles / species for moles	(2)

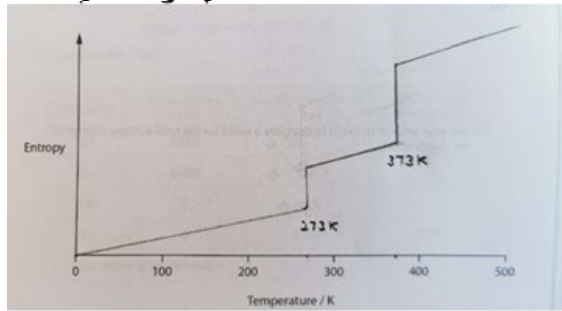
Question Number	Answer	Additional Guidance	Mark
19(b)(i)	• calculation of sum of entropy of products • calculation of sum of entropy of reactants • calculation of entropy of the system	(1) <u>Example of calculation</u> $\sum S^\ominus_{(products)} = ((2 \times 192.3) + (2 \times 69.9) + 213.8 =)$ $(+)738.2 \text{ (J K}^{-1} \text{ mol}^{-1}\text{)}$ (1) $\sum S^\ominus_{(reactants)} = ((2 \times 151.1) + 99.7 =)$ $= (+)401.9 \text{ (J K}^{-1} \text{ mol}^{-1}\text{)}$ (1) $\Delta S^\ominus_{system} = (738.2 - 401.9 =)$ $= (+)336.3 \text{ (J K}^{-1} \text{ mol}^{-1}\text{)}$ Ignore SF except 1SF Ignore units, even if incorrect Correct answer with no working scores (3) TE throughout	(3)

Question Number	Answer	Additional Guidance	Mark
19(b)(ii)	• calculation of sum of enthalpy change of formation of products (1) • calculation of sum of enthalpy change of formation of reactants (1) • calculation of enthalpy change of reaction (1)	<u>Example of calculation</u> $\sum \Delta H^\circ_{(products)} = ((2 \times -285.8) + (2 \times -46.1) + -992.1 =)$ $= -1655.9 \text{ (kJ mol}^{-1}\text{)}$ $\sum \Delta H^\circ_{(reactants)} = ((2 \times -365.6) + -944.7 =)$ $= -1675.9 \text{ (kJ mol}^{-1}\text{)}$ $\Delta_r H^\circ = (-1655.9 - -1675.9 =)$ $= (+) 20.0 \text{ (kJ mol}^{-1}\text{)}$ Ignore SF except 1SF Ignore units, even if incorrect, Correct answer with no working scores (3) TE throughout -20 (kJ mol⁻¹) scores (2)	(3)

Question Number	Answer	Additional Guidance	Mark
19(b)(iii)	- calculation of entropy of the surroundings (1) - calculation of total entropy change with sign and units (1)	<u>Example of calculation</u> $\Delta S^{\circ}_{\text{surroundings}} = (-\Delta H \div T) = (-20\,000 \div 298)$ $= -67.1141 / -67.1 \text{ (J K}^{-1} \text{ mol}^{-1}\text{)}$ $\Delta S^{\circ}_{\text{total}} = \Delta S^{\circ}_{\text{system}} + \Delta S^{\circ}_{\text{surroundings}}$ $\Delta S^{\circ}_{\text{total}} = (+336.3 - 67.1)$ $= +269.2 \text{ J K}^{-1} \text{ mol}^{-1}$ Allow +0.269 kJ K⁻¹ mol⁻¹ Ignore SF except 1SF Correct answer with no working scores (2) TE throughout except for negative values of $\Delta S^{\circ}_{\text{total}}$	(2)

Question Number	Answer	Additional Guidance	Mark
19(c)(i)	An answer that makes reference to the following points: (molecules of) chlorine disperse/diffuse into the top jar (1) - they have greater entropy (because they are more spread out/ random) (1)	Allow chlorine mixes with the air (in the top jar) Allow chlorine moves into the top jar Allow disorder / randomness for entropy Ignore the molecules / chlorine is more dispersed for M2	(2)

Question Number	Answer	Additional Guidance	Mark												
19(c)(ii)	An answer that makes reference to the following points: correct completion of bottom three rows	<table><tr><th>Molecule A</th><th>Molecule B</th></tr><tr><td>2</td><td>2</td></tr><tr><td>3</td><td>1</td></tr><tr><td>1</td><td>3</td></tr><tr><td>4</td><td>0</td></tr><tr><td>0</td><td>4</td></tr></table> Accept rows in any order	Molecule A	Molecule B	2	2	3	1	1	3	4	0	0	4	(1)
Molecule A	Molecule B														
2	2														
3	1														
1	3														
4	0														
0	4														

Question Number	Answer	Additional Guidance	Mark
19(d)	- general shape of increase from left to right - two vertical stages for melting and boiling - include the use of 273 for melting and 373 for boiling either by labelling or position on x axis	(1) Allow horizontal sections between phase changes Allow a straight / curved line (1) Ignore height of lines (1) <u>Example of graph</u> 	(3)

Question Number	Answer	Additional Guidance	Mark
19(e)	An answer that makes reference to two of the following points: • in the solid/lattice the ions are fixed (1) • but in solution the ions are mobile (so more dispersed) (1) • and the hydrogen bonds between molecules of water are disrupted (1)	Allow there are more ways of arranging the ions / energy quanta Allow particles for ions in M2 only If no other mark awarded, allow (1) for an increase in the number of species / particles Do not award reference to molecules other than water Do not award increase in the number of ions Penalise reference to atoms once only	(3)

(Total for Question 19 = 21 marks)
TOTAL FOR SECTION C = 21 MARKS
TOTAL FOR PAPER = 90 MARKS